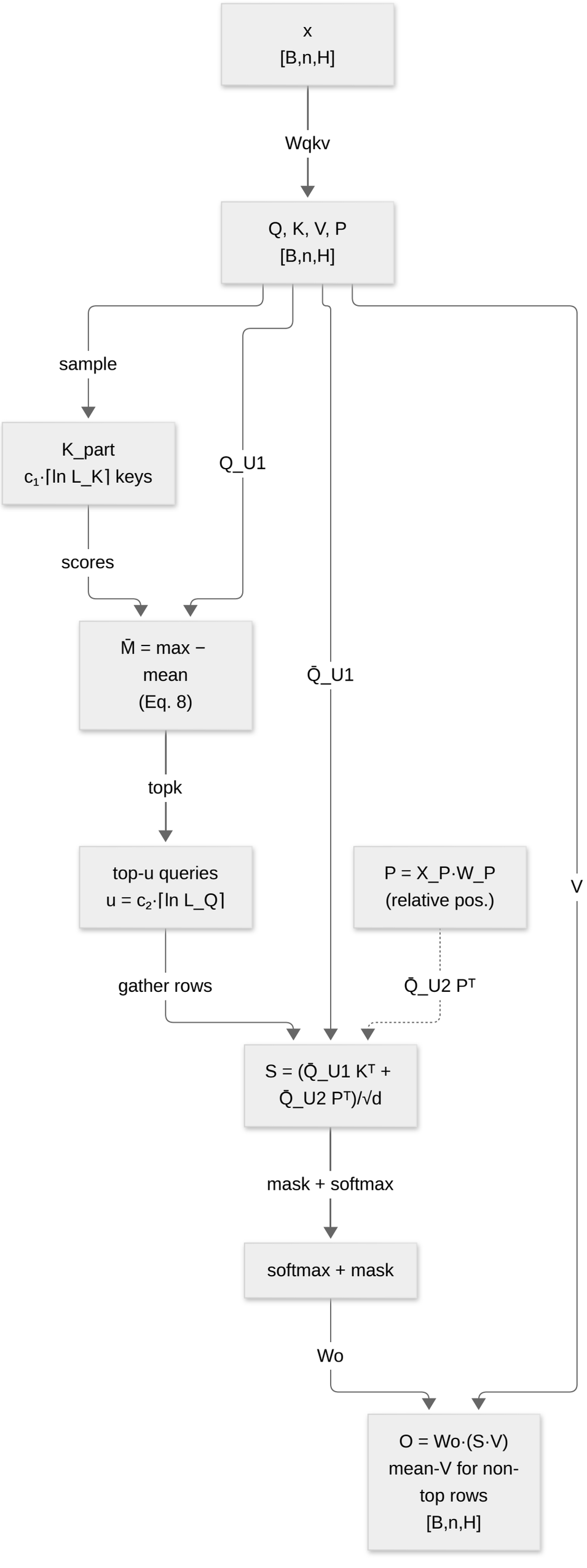

x
[B,n,H]
Wqkv
Q, K, V, P
[B,n,H]
sample
K_part
c₁·⌈ln L_K⌉ keys
Q_U1
scores
M̄ = max − mean
(Eq. 8)
Q̄_U1
topk
top-u queries
u = c₂·⌈ln L_Q⌉
P = X_P·W_P
(relative pos.)
V
gather rows
Q̄_U2 Pᵀ
S = (Q̄_U1 Kᵀ + Q̄_U2 Pᵀ)/√d
mask + softmax
softmax + mask
Wo
O = Wo·(S·V)
mean-V for non-top rows
[B,n,H]